



RESEARCH ARTICLE

Assessment of Water Quality for Drinking from Selected Places of Nagapattinam City using Water Quality Index

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ABSTRACT

This paper is assessed the quality of the ground water resources from Nagapattinam city, based on drinking water quality index namely arithmetic and geometric one. Separate containers have been used to collect water from selected places of Nagapattinam and examined for electrical conductivity, pH, Cl⁻ ion, SO₄²⁻ ion, total alkalinity, total hardness, Ca²⁺ ion, Mg²⁺ ion, presence of heavy metal iron, F⁻ ion and NO₃⁻ ions. World Health Organization(WHO) and Bureau of Indian Standards (BIS) recommended values were made comparison with the obtained results. Arithmetic and geometric drinking water quality index values were calculated for all the samples. From the result, one can get a conclusion that arithmetic drinking water quality index categorized the same samples as “poor” category due to the occurrence of eclipsing phenomena. But geometric drinking water quality assessed the same sample as “very poor”. Besides both the drinking water quality indices reported that all analyzed water samples are not in the level of using as drinking water directly. But geometric water quality index picked up the original

105750



**Ponsadai Lakshmi et al.,**

quality of water samples. Therefore, an immediate call on water managers for putting a separate mechanism to treat the ground water resources before using for drinking purposes.

Keywords: Ground water, Nagapattinam, arithmetic drinking water quality index, geometric drinking water quality index, eclipsing phenomena

INTRODUCTION

Quality of drinking water samples can be judged by the presence of various constituents originated from different sources namely physical origin, chemical source and biological ways. The presence of various ions inside the permissible limit make the water sample suitable for drinking purpose. Presence of various ions in various levels, suppose if it crossed the limiting value make it complicated to get the final quality which in turn will reflect in the quality. Hence decision making is important. Human beings are affected by many diseases, out of which 80% and above are directly related to water purity nature, sources from World Health Organization [1]. Water professionals told that the original nature of samples can be received by making direct comparison of the result obtained from the analytical part with recommended values, this step is purely technical based one, does not give the original nature of samples used for drinking [2]. To reduce this complication for getting accurate decision,[3] proposed WQI -Water Quality Index. This WQI described the purity of water samples in a gentle way. Many corrections have been made on this to improve its quality, recommended by using the delphi technique recommended by National Sanitation Foundation (NSF), USA. WQI is a mathematical based procedure, which combines results of all parameters analytical data single numeral. From this the overall quality of water can be known. According to [4], [5] many advances have been made to modify the basic principles WQI. Scientists such as [6], [7], [8], [9], [10], [11], [12] were also helped a lot to make changes on WQI. Different way of developing sub index and their aggregation methods were the main tools during advancement of WQI. Finally overall quality of different water samples collected from different environment was clearly given by both WQIs. Hence, to judge the overall water quality of samples, suitable DWQI should be identified to make the samples suitability for drinking purpose.

MATERIALS

Sampling and analysis of samples

10 sites were selected for sampling of ground water from Nagapattinam city. The details of places from which water samples were collected and analyzed, represented in Fig.1 and tabulated in Table 1. Standard methods were used for sample collection and the analysis of different parameters such as electrical conductivity, pH, Cl^- ion, SO_4^{2-} ion, total alkalinity, total hardness, Ca^{2+} ion, Mg^{2+} ion, presence of heavy metal iron, F^- ion and NO_3^- ions were carried out using standard procedures given by [13].

Description of Water Quality Index value for drinking

It is mathematical procedure by which the conversion of all the obtained results of various physical, chemical constituents with different units can be converted into a single numeral. From this numeral, the real quality of the sample can be obtained. Two stages were used to get value of DWQI, The first one is the creation of unitless subindex values from various physical, chemical constituents with different units. In step number two, the aggregation of sub index value using some functions into water quality index value.

Selection of Parameters

Parameters have been selected with different weightages based on its impact on health related value. In one way their presence is most important, the level should not exceed the prescribed limit. If it exceeds many harmful effects will be received by human beings [14].





Ponsadai Lakshmi et al.,

Considering the above points, 11 parameters were selected for assessing the quality for drinking. Water quality constituents namely EC, pH, chloride ion, sulphate ion, total alkalinity, total hardness, Ca^{2+} ion, Mg^{2+} ion, heavy metal iron and highly health oriented F^- ion, NO_3^- ions were taken into consideration for calculating indices.

Creation of Sub-Index (SI)

In WQI concept different units physical and chemical constituents with different dimensions are converted into unitless sub-index value, bringing the different units and dimensions analytical results into common scale. Here each water quality constituent was given with values starting from 0 to 100 as rating value during development of sub index value. Based on guideline values of individual parameters, the rating value has been assigned to each parameter which was recommended by [15], [16] and [17] in 2nd Table. If SI (sub index) with rating = 100 means that the sample is within the desirable limit. But sub-index with rating = 50 means that the sample is reached the maximum acceptable limit value. Based on regression statistics, samples quality fell between zero to hundred [2] and [18]. Electrical conductivity, pH, chloride ion, sulphate ion, total alkalinity, total hardness, Ca^{2+} ion, Mg^{2+} ion, heavy metal iron, F^- ion and NO_3^- ions.

Allocation of temporary weight to the constituents

Initially, a temporary weight (W_i) is given to every physical and chemical constituents. This weight allocation is purely depends upon the percentage of weightage of that parameter for drinking. Divide the each parameters temporary weight by the addition of all of temporary weights [Eq. 1].

equation 1 should be like this $W = W_i$ (1)

$$\sum_{i=1}^n (W_i)$$

Calculation of DWQI

DWQI is calculated by aggregating weight factors and values of subindex of each water quality constituents, with the help of well defined aggregation function. Usually addition and multiplication aggregation techniques are the famous ones.

In this research study, the obtained SI values were combined by weighted additive and multiplicative methods

- (i) Value of drinking water quality index by addition or arithmetic ($\text{DWQI}_{(A)}$) was calculated by involving [2].

equation 2 should be like this $\text{DWQI}_{(A)} = \sum_{i=1}^n [SI_i \times W_i]$ (2)

- (ii) Value of drinking water quality index by multiplication or geometric ($\text{DWQI}_{(G)}$) was calculated by involving [2]

$$\text{DWQI}_{(G)} = \prod_{i=1}^n SI_i^{W_i} \text{ (3)}$$

here,

SI_i – ith constituents sub index value

W_i – ith constituents weight factor value

n – number of constituents involved ($n = 11$)

Using the above equation Weight factors of all water quality parameters can be calculated. For this calculation temporarily assigned weight (W_i) and the total numbers of constituent (n) were used and $n=11$ here (Table 3).



**Ponsadai Lakshmi et al.,****Categorization of Scores of Drinking Water Quality Index**

Equations with aggregating scores were developed from the rating values 0 and 100. Zero represents worst quality of water and hundred represents excellent quality of water. Drinking water quality index scores with its detailed classification is presented in 4th Table.

RESULTS AND DISCUSSION**Study area quality assessment through DWQI values for drinking**

From Nagapattinam city, 10 different places were identified for sample collection and for all the samples DWQI value has been calculated. To calculate DWQI addition and multiplication WQI methods were used, the values were tabulated in the 5th Table. The scores of arithmetic DWQI is greater than the scores of geometric DWQI was noted during the calculation. Low values were received for geometric DWQI. From the results, it was observed that AWQI scores dominated GWQI for all the samples. Arithmetic drinking water quality indicated the quality was 'POOR' for all samples. But geometric drinking water quality indexing system indicated 'VERY POOR' for all the samples. From these two DWQI calculation, it was noticed that arithmetic DWQI gave 'POOR' quality category, geometric DWQI reported the status of 'VERY POOR' for all the analyzed samples. From the insight analysis of water quality parameters reported that parameters like total alkalinity of sample-5, total hardness of samples-10, chloride value of samples -3,4,5 and 10, magnesium value of -5 were occasionally crossed the maximum acceptable limit of drinking water quality of guideline value. This occasional crossing of maximum permissible limit of guideline values of various parameters such as total hardness, total alkalinity which have high impact on health impact. It was reflected in the result geometric DWQI while assessing the drinking water quality of water samples. That is why geometric DWQI assessed the samples were in 'VERY POOR' category (Fig.2). [19] and [20] proposed the following views that is arithmetic and geometric drinking water quality indices gave two different results for the same water samples. This was in agreement with the views of [21], [22], [23], [24], [25] and [26]. The above researchers told that the eclipsing problem may happen when the used aggregation method is addition. When at least one parameter was not up to the supportive level of drinking water quality as per guideline value, eclipse may arise. So arithmetic DWQI was forced to underestimate assess status of estimating the pollution nature. So arithmetic DWQI is unable to appraise the real drinking quality of water and this reflects in the judgement of arithmetic DWQI. Hence, arithmetic DWQI over estimated the quality. If more number of physical and chemical constituents chosen for calculating WQI, the value of SI value is lower and delicate while aggregating values. That is DWQI arithmetic over judged the original nature of samples than DWQI geometric.

CONCLUSION

In this study, crystal clear judgement was given by geometric DWQ Indexing system when compared to arithmetic DWQ Indexing system. This is because of eclipsing phenomena which in turn reflected in the judgement of Arithmetic DWQI. So it could not tell the nature of ground water sources accurately. But the real impairment of samples were identified by geometric DWQI. Decision of geometric DWQI helped to start the control action towards minimizing pollution. In nutshell, geometric DWQ Indexing assessed the samples in transparent manner, water needs to be purified before consumption.

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Ponsadai Lakshmi et al.,

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Table.1: Name of the sampling locations

Sl.No	Sampling Locations
S-1	No - 5, Sebastiyarkovil Street, Kadambadi
S-2	No - 6, Sebastiyarkovil Street, Kadambadi
S-3	12, Pilliarkovil street, Kadambadi
S-4	22, GH Road, velippalayam
S-5	Sevabharathi, Tata nagar
S-6	47, Saveriyarkovil street, Kadambadi
S-7	23/2, Mahalakshminagar, Kadambadi
S-8	1 st lane beach road,
S-9	Teachers Avenue, 1 st lane beach road
S-10	38, Sebastiyarkovil street, kadambadi

Table 2: Values recommended for checking quality of drinking water

Characteristics with chemical constituents in mg/l	Bureau of Indian Standard (2003)		World Health Organization (2006b)	
	Desirable	Max. allowed	Allowed	Max. allowed
pH	6.5 – 8.5	No relax.	6.5 – 8.5	6.5 – 9.2
EC (dS/m)	1	3	-	-
Total Alkalinity (in terms of CaCO ₃ equivalent)	200	600	500	-
Total hardness (in terms of CaCO ₃ equivalent)	300	600	200	500
Calcium	75	200	-	-
Magnesium	30	100	-	-
Chloride	250	1000	250	600
Fluoride	1.0	1.5	-	1.5
Sulphate	200	400	250	500
Nitrate	45	No relax.	-	50
Iron	0.3	1.0	0.3	1.0 – 3.0

Table.3: Assigning temporary weight for individual physical and chemical constituents

Physical and chemical constituents	Temporary Weights (W _t)	Weight factor (W)
		Conventional DWQI
pH	4.5	0.037
EC	5.0	0.041
Chloride	5.0	0.041
Sulphate	5.5	0.045
Total Alkalinity	4.5	0.037
Total Hardness	4.5	0.037
Calcium	4.5	0.037
Magnesium	4.5	0.037
Iron	6.0	0.050
Fluoride	9.0	0.074
Nitrate	10.0	0.083

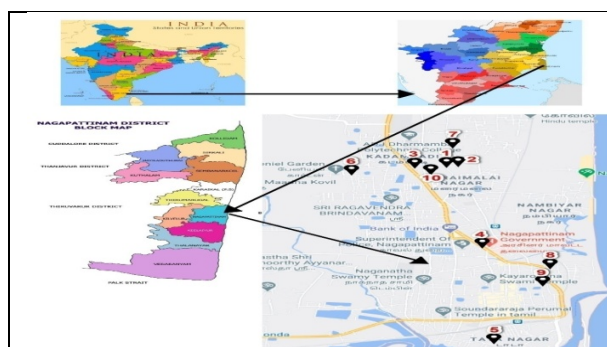
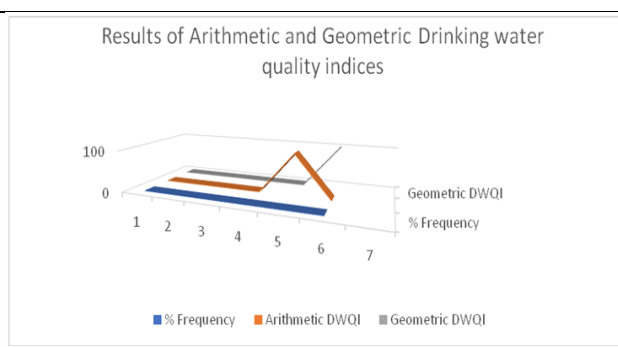


**Table 4: Categorization drinking water quality index scores**

Indices Score range	Category	Remarks
≥ 95.0 to 100	Excellent	'Best Quality' Water samples were within the desirable levels prescribed in guidelines
≥ 85.0 to < 95.0	Good	'Good Quality' Occasionally water samples crossing the guideline values.
≥ 75.0 to < 85.0	Fair	'Acceptable Quality' Water samples are frequently exceeding the guideline values. But deviation from quality is very minimum.
≥ 60.0 to < 75.0	Marginal	'Threatened Quality' Some times exceeding from acceptable guideline values.
≥ 40.0 to < 60.0	Poor	'Poor Quality' Exceeding the acceptable levels of guideline values.
< 40.0	Very Poor	'Worst Quality' Always exceeding the acceptable levels of Guideline values.

Table 5: Additive and Multiplicative water quality index values for the analyzed sample

Sl.No	Locations	Additive DWQI	Quality	Multiplicative DWQI	Quality
S-1	No - 5, Sebastiyarkovil Street, Kadambadi	53.7	Poor	11.1	Very Poor
S-2	No - 6, Sebastiyarkovil Street, Kadambadi	54.3	Poor	11.2	Very Poor
S-3	12, Pilliarkovil street, Kadambadi	52.3	Poor	11.0	Very Poor
S-4	22, GH Road, velippalayam	52.3	Poor	11.0	Very Poor
S-5	Sevabharathi, Tata nagar	49.3	Poor	10.5	Very Poor
S-6	47, Saveriyarkovil street, Kadambadi	51.7	Poor	10.9	Very Poor
S-7	23/2, Mahalakshminagar, Kadambadi	41.4	Poor	8.4	Very Poor
S-8	1 st lane beach road,	51.2	Poor	10.8	Very Poor
S-9	Teachers Avenue, 1 st lane beach road	53.7	Poor	11.1	Very Poor
S-10	38, Sebastiyarkovil street, kadambadi	50.1	Poor	10.6	Very Poor

**Fig.1: Figural representation of sampling stations****Fig. 2: Results of Drinking Water Quality Indices in the ground water samples of the study area**